Elements of Decision Theory: Part II

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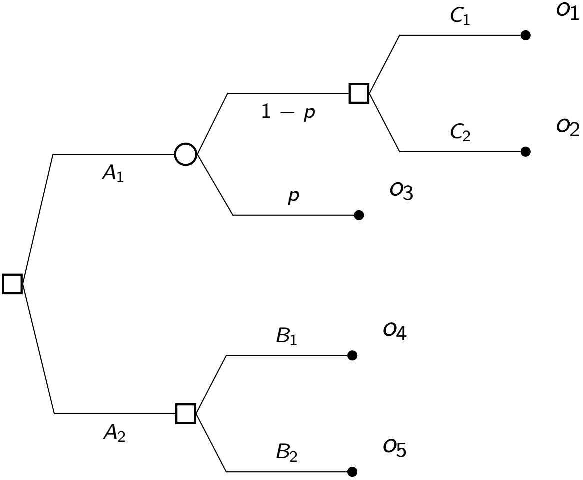
Sequential decision making: decision tree representation

In a ***sequential decision problem*** an individual is called to take a temporal ordered sequence of structurally dependent decisions.

A ***decision tree*** is a tree in which there are:

- *decision* (or choice) *nodes* (represented by $\square$) at which decisions among risky alternatives are taken;
- *event* (or chance) *nodes* (represented by $\bigcirc$) at which an event occurs with some given (objective) probability;
- *terminal nodes* (represented by $\bullet$) that are the leaves of the tree and display the final utility of some path from the root to the corresponding leaf;
- *lines* from decision nodes to event nodes or terminal nodes represent choices, lines from event nodes to decision nodes or terminal nodes represent stochastic outcomes, i.e. realizations of the possible states of the world.

Sequential decision making: decision tree representation



Sequential decision making: rationality over time

In sequential decision problems, it is common to identify three main *behavioral* approaches:

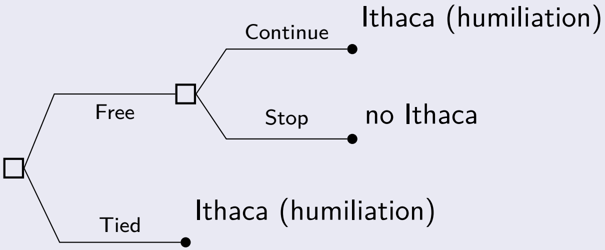
MYOPIC/NAIVE ATTITUDE – the decision maker myopically faces **each stage independently** (i.e. optimizing one-stage utility greedily, not evaluating the impact of current choices on future outcomes).

RESOLUTE ATTITUDE – the decision maker **commits to** some predetermined **plan** (i.e. even though they might be willing to deviate from it at some point in the future, they stick to the plan independently of future outcomes).

SOPHISTICATED ATTITUDE – the decision maker elaborates an *optimal plan* **anticipating** their future choices (i.e. they adopt the best plan among those they are willing/capable to follow till the end).

Sequential decision making: rationality over time

Example



- *Myopic attitude*: (Free, No Ithaca).
- *Resolute attitude*: (Free, Continue)
- *Sophisticated attitude*: (Tied, Ithaca)

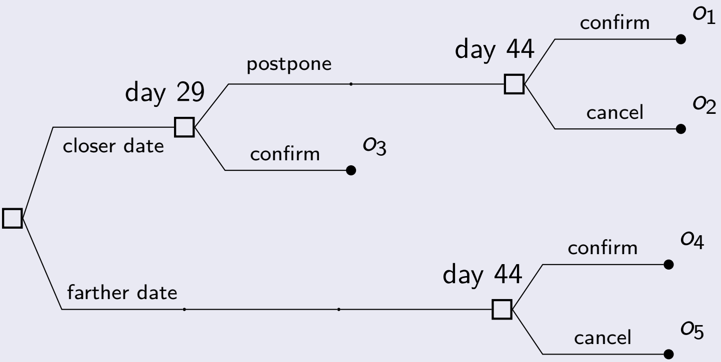
Sequential decision making: rationality over time

Example (Exam Planning)

Bob is planning the date of an exam. The teacher proposes two dates: 30 or 45 days from now; Bob must decide which one to pick. In the first case, at day 29, Bob could still confirm his presence the next day or postpone it to the farther date. On day 44, instead, Bob can just confirm the presence the next day or cancel it al all.
What should Bob decide?

Sequential decision making: rationality over time

Example (cont'd)



Suppose $u(o_3) > u(o_4) > u(o_1) > u(o_5) > u(o_2)$

Rational agents (according to EUT) choose "closer date & confirm".

Example (cont'd)

Assume *geometric time discounting*

$$U_t(x_t, x_{t+1}, x_{t+2}, \dots) = u(x_t) + \delta u(x_{t+1}) + \delta^2 u(x_{t+2}) + \dots$$

- instantaneous pain due to decision of taking exam (-10)
- reward for passing the exam on the exam day ($+25$)
- canceling the exam (0)
- preference for closer date (ϵ)
- instantaneous relief for postponing ($+1$)
- later regret for postponing (-2)
- daily temporal discount $\delta = 0.99$.

Example (cont'd)

The **resolute** agent chooses the path that maximizes the overall utility, as seen from the initial day perspective

$$
\begin{aligned}
U_0(o_3) &= \epsilon + \delta^{29}(-10) + \delta^{30}(+25) \approx 11 \\
U_0(o_4) &= 0 + \delta^{44}(-10) + \delta^{45}(25) \approx 9.5 \\
U_0(o_1) &= \epsilon + \delta^{29}(+1) + \delta^{44}(-12) + \delta^{45}(25) \approx 8.9 \\
U_0(o_2) &= \epsilon + \delta^{29}(+1) + \delta^{44}(0) \approx 0.75 \\
U_0(o_5) &= 0 + \delta^{44}(0) = 0
\end{aligned}
$$

The resolute agent chooses the closer date, then confirms.

Example (cont'd)

The **myopic** agent, at each node, chooses the action that maximizes the local utility (as seen from the perspective of that day)

$$\begin{aligned} U_0\,(\text{closer}) &= \epsilon \\ U_0\,(\text{farther}) &= 0 \\ U_{29}\,(\text{confirm}) &= -10 \\ U_{29}\,(\text{postpone}) &= 1 \\ U_{44}\,(\text{confirm}) &= -10 \\ U_{44}\,(\text{cancel}) &= 0 \end{aligned}$$

The myopic agent chooses the closer date, then postpones.
To obtain the same result local preferences have to align with global ones at every node.

Example (cont'd)

Here we assumed being myopic as only caring for immediate reward/costs. If we relax this definition, assuming that the agent, at each node, only cares about the utility directly related to the decision taken at that node, we will obtain

$$
\begin{aligned}
U_0\left(\text{closer}\right) &= \epsilon \\
U_0\left(\text{farther}\right) &= 0 \\
U_{29}\left(\text{confirm}\right) &= -10+\delta\left(25\right)\approx 14.75 \\
U_{29}\left(\text{postpone}\right) &= +1+\delta^{15}\left(-2\right)\approx -0.72 \\
U_{44}\left(\text{confirm}\right) &= -10+\delta\left(25\right)\approx 14.75 \\
U_{44}\left(\text{cancel}\right) &= 0
\end{aligned}
$$

The (mildly) myopic agent would then choose the closer date and confirm.

Example (cont'd)

The **sophisticated** agent, at each node, maximizes on the continuation path (as seen from the perspective of that day)

$$\begin{aligned}
U_{44}\left(\text{confirm}\right) &= -10+\delta 25\approx 14.75\\
U_{44}\left(\text{cancel}\right) &= 0\\
U_{29}\left(\text{confirm}\right) &= -10+\delta\left(25\right)\approx 14.75\\
U_{29}\left(\text{postpone}\right) &= +1+\delta^{15}\left(-12\right)+\delta^{16}\left(25\right)\approx 11.97\\
U_{0}\left(\text{closer}\right) &= \delta^{29}\left(-10\right)+\delta^{30}\left(25\right)\approx 11\\
U_{0}\left(\text{farther}\right) &= 1+\delta^{44}\left(-12\right)+\delta^{45}\left(25\right)\approx 9.2
\end{aligned}$$

The sophisticated agent chooses the closer date, then confirms. Maximizing over subpaths leads to global maximization because the utility is separable (time consistent).

Sequential decision making: rationality over time

Definition

A sequential decision problem is **sequentially** (dynamically) **consistent** if the way an agent evaluates their future choices depends only on the reachable outcomes from that node forward.

The corresponding utility function is **separable**.

Definition

A sequential decision problem satisfies **time consistency** if a plan which is considered optimal today is still considered optimal when any future decision point within that plan is reached.

Remark

Separability implies time consistency. Non separability can lead to time inconsistency.

Sequential decision making: rationality over time

Example (Exam Planning with time inconsistency)

All future rewards are penalized compared to present ones (*present bias*)

$$U_{\mathrm{t}}\left(x_t, x_{t+1}, x_{t+2}, \dots\right) = u\left(x_t\right) + \beta\left[\delta u\left(x_{t+1}\right) + \delta^2 u\left(x_{t+2}\right) + \dots\right]$$

- same rewards and costs
- $\beta = 0.75$ and $\delta = 1$

Other common form of time inconsistency: **hyperbolic discounting**

$$U_{\mathrm{t}}\left(x_t, x_{t+1}, x_{t+2}, \dots\right) = \sum_{\tau \geq t} \frac{1}{1 + \alpha\left(\tau - t\right)} u\left(x_\tau\right)$$

Example (cont'd)

The **resolute** agent chooses the path that maximizes the overall utility, as seen from the initial day perspective

$$
\begin{aligned}
U_0(o_3) &= \epsilon + \beta(-10 + 25) \approx 11.25 + \epsilon \\
U_0(o_4) &= \beta(-10 + 25) \approx 11.25 \\
U_0(o_1) &= \epsilon + \beta(+1 - 12 + 25) \approx 10.5 \\
U_0(o_2) &= \epsilon + \beta(+1 - 2) = -\beta \\
U_0(o_5) &= 0
\end{aligned}
$$

The resolute agent chooses the closer date, then confirms (CC).

Sequential decision making: rationality over time

Example (cont'd)

The **myopic** agent, at each node, chooses the action that maximizes the local utility (as seen from the perspective of that day)

$$
\begin{aligned}
U_0\,(\text{closer}) &= \epsilon \\
U_0\,(\text{farther}) &= 0 \\
U_{29}\,(\text{confirm}) &= -10 \\
U_{29}\,(\text{postpone}) &= 1 \\
U_{44}\,(\text{confirm}) &= -10 \\
U_{44}\,(\text{cancel}) &= 0
\end{aligned}
$$

The myopic agent chooses the closer date, but then postpones. A mildly myopic agent would instead choose closer and confirm $[-10 + \beta\,(25) \approx 8.75 > 1 - 2\beta = -0.5]$.

Sequential decision making: rationality over time

Example (cont'd)

The **sophisticated** agent, at each node, maximizes on the continuation path (as seen from the perspective of that day)

$$\begin{aligned} U_{44}\,(\text{confirm}) &= -10 + \beta\,(25) \approx 8.75 \\ U_{44}\,(\text{cancel}) &= 0 \\ U_{29}\,(\text{confirm}) &= -10 + \beta\,(25) \approx 8.75 \\ U_{29}\,(\text{postpone}) &= 1 + \beta\,(-12 + 25) \approx 10.75 \\ U_0\,(\text{closer}) &= \epsilon + \beta\,(1 - 12 + 25) \approx 10.5 \\ U_0\,(\text{farther}) &= \beta\,(-10 + 25) \approx 11.75 \end{aligned}$$

The sophisticated agent anticipates that on day 29 they will prefer postponing, therefore chooses the farther date, then confirms (FC).

Sequential decision making: rationality over time

Example (cont'd)

In summary:

- *Myopic attitude* (farther date, confirm): decisions are taken independently, the student first prefers the closer date, then prefers to postpone it to increase the chance of passing the exam.
- *Resolute attitude* (closer date, confirm): the student does not anticipate the future state of the world, but tries to stick to the prearranged plan even if unfeasible/unfavorable. Re-evaluation of plans might lead to *sequential inconsistency,* which is overridden by willpower and commitment.
- *Sophisticated attitude* (farther date, confirm): The student anticipates he won't resist to the opportunity of postponing the date of the exam, he prefers to set it on the farther date from the beginning.

Sequential decision making: rationality over time

The sophisticated attitude (**optimal planning**) always guarantees ***sequential consistency*** of preferences, but it requires the ***separability*** of the expected utility to also find global optimum (*optimal substructure*).

Remark

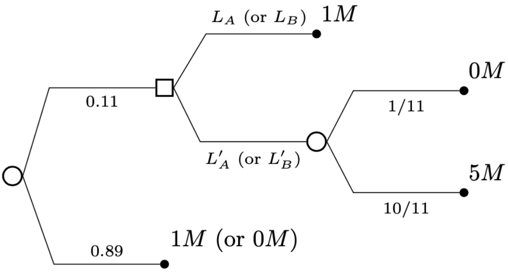
The separability property is essentially the **independence axiom** of the EUT **applied across time**.

Independent axiom: agent's preference between lotteries should be independent of a common alternative lottery.

In a decision tree, the common alternative is the history that led to the decision node; while lotteries are the future continuation paths.

Sequential decision making: rationality over time

Allais' paradox can be made sequential.



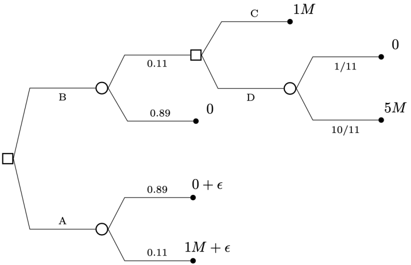
The paradox $L_A \succsim L'_A$ and $L'_B \succsim L_B$ is a signature of sequential inconsistency because preference at the decision node are reversed.

Remark

The independence axiom implies the separability principle.

A modified Allais sequential decision problem

The value ϵ is such that $A \succsim BC$, but $BD \succsim A$.
However, after Allais we should assume $C \succsim D$.
Here sequential inconsistency induces to choose a *dominated* action (A is always better than C).



Remark

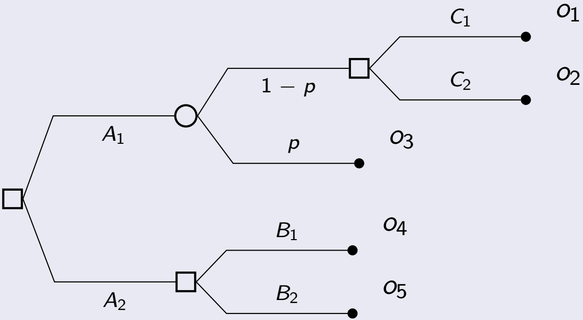
Resolute attitude can avoid sequential inconsistency by relaxing the separability principle. Sophisticated agents avoid sequential inconsistency within the assumption of the separability principle.

Sequential decision making: backward induction

Sophisticated agents apply the principle of (expected) utility maximization to sequential decisions assuming separability (optimal substructure).

The process is called ***backward induction***: start from the leaves and perform maximization at choice nodes and expectation at chance nodes.

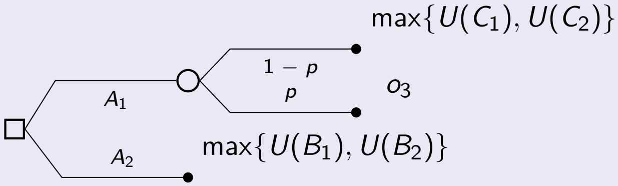
Example



Example (cont'd)

Given $U(C_1) = u(o_1), U(C_2) = u(o_2), U(B_1) = u(o_4)$ and $U(B_2) = u(o_5)$,

$$
\begin{aligned}
U(A_1) &= p u(o_3) + (1-p) \max\left\{U(C_1), U(C_2)\right\}, \\
U(A_2) &= \max\left\{U(B_1), U(B_2)\right\}.
\end{aligned}
$$



Finally, we compute $A_{\text{opt}} = \arg\max\left\{U(A_1), U(A_2)\right\}$.

Sequential decision making: information processing

In sequential decision problems, the uncertainty about the state of the world can be quantified by means of ***Bayesian reasoning***.

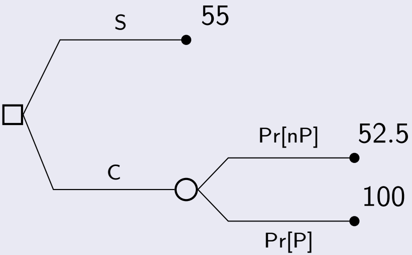
Example

John is deciding whether to invest 50 kEur in a company's shares (C). There are rumors that the company will sell stock shares publicly within the year. If so, he could double his money, otherwise he will get just 5% return per year. As an alternative, John could buy saving certificates (S) at 10% return per year.

	company goes public (P)	don't go public (nP)
invest in the company (C)	100	52.5
invest in certificates (S)	55	55

Suppose the a priori probability p of going public is known, what is the best alternative available to John?

Example (cont'd)



Maximize the expected utility given the prior information, then

$$
\begin{aligned}
\mathbb{E}\left[u\left(\mathrm{C}\right)\right] &= p100 + (1-p)\,52.5 = 47.5p + 52.5, \\
\mathbb{E}\left[u\left(\mathrm{S}\right)\right] &= p55 + (1-p)\,55 = 55.
\end{aligned}
$$

Choice C is rational if $47.5p + 52.5 > 55$, i.e. $p > 2.5/47.5 \approx 0.053$.

Sequential decision making: information processing

Example

John can learn additional information (from an insider) sufficient to discriminate whether the company will go public or not. How much is he willing to pay such information?

John's *strategy* σ: buy information (B), then if the company goes public for sure, then play C, otherwise S.

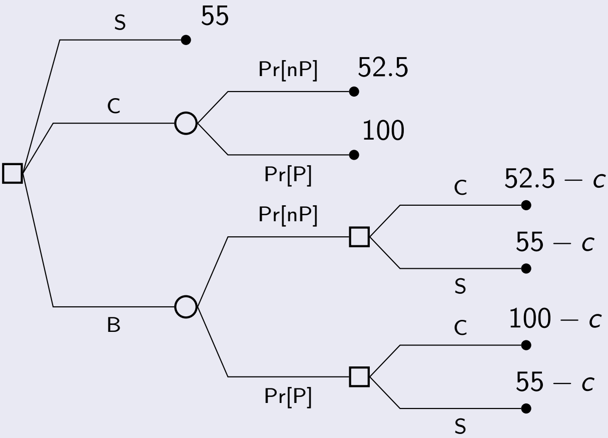
$$U(\sigma) = \mathbb{E}[u(\sigma)] = pu(\mathrm{C}) + (1-p)\, u(\mathrm{S}) = 45p + 55$$

The *value of additional information* is the marginal gain

$$\mathbb{E}[u(\sigma)] - \mathbb{E}[u(\mathrm{C})] = 2.5(1-p) > 0$$

The insider can sell the information at a cost $c \leq 2.5(1-p)$.

Example (cont'd)



Is σ the optimal strategy? Check it by *backward induction*.

Sequential decision making: information processing

Example

Suppose the insider provides an internal report containing *imperfect information*: report's outcome is Y (resp. N) if public offer is reported (resp. denied). We assume that

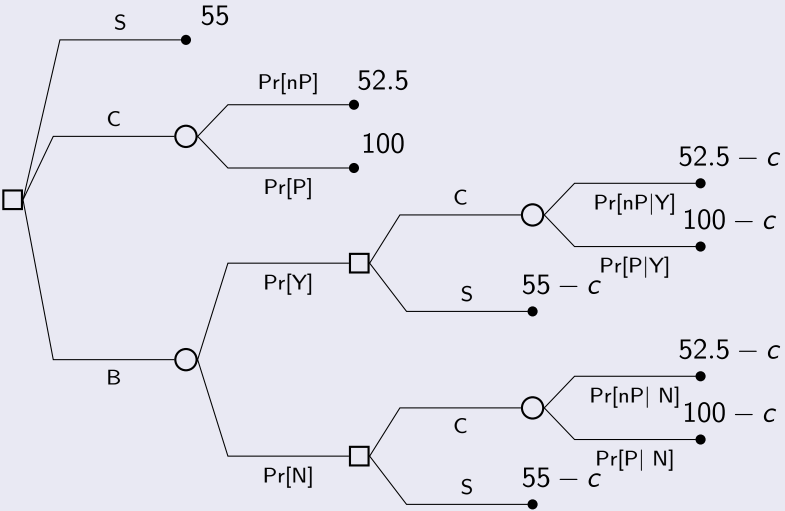
- if the company goes public (P) there is a probability $q_1 = \Pr[\mathrm{Y}|\mathrm{P}]$ that it is revealed in the report,
- if the company is not going public (nP) there is a probability $q_2 = \Pr[\mathrm{N}|\mathrm{nP}]$ that it is said so in the report.

John can:

- directly play S (certain outcome)
- directly play C (subject to prob p)
- bribe the insider (B) and postpone the decision after report reading

Apply *Bayesian reasoning* to adjust probabilities in the sequential process.

Example (cont'd)



Sequential decision making: information processing

Example (cont'd)

Applying *Bayes rule* (assume $p = 0.5, q_1 = 0.9, q_2 = 0.5$):

$$\begin{aligned}
\Pr[\mathrm{P}|\mathrm{Y}] &= \frac{\Pr[\mathrm{Y}|\mathrm{P}]\Pr[\mathrm{P}]}{\Pr[\mathrm{Y}]} = \frac{\Pr[\mathrm{Y}|\mathrm{P}]\Pr[\mathrm{P}]}{\Pr[\mathrm{Y}|\mathrm{P}]\Pr[\mathrm{P}] + \Pr[\mathrm{Y}|\mathrm{nP}]\Pr[\mathrm{nP}]} \\
&= \frac{q_1 p}{q_1 p + (1-q_2)(1-p)} = 0.64
\end{aligned}$$

$$\begin{aligned}
\Pr[\mathrm{P}|\mathrm{N}] &= \frac{\Pr[\mathrm{N}|\mathrm{P}]\Pr[\mathrm{P}]}{\Pr[\mathrm{N}]} = \frac{\Pr[\mathrm{N}|\mathrm{P}]\Pr[\mathrm{P}]}{\Pr[\mathrm{N}|\mathrm{P}]\Pr[\mathrm{P}] + \Pr[\mathrm{N}|\mathrm{nP}]\Pr[\mathrm{nP}]} \\
&= \frac{(1-q_1)p}{(1-q_1)p + q_2(1-p)} = 0.17
\end{aligned}$$

Then $\Pr[\mathrm{nP}|\mathrm{Y}] = 1 - \Pr[\mathrm{P}|\mathrm{Y}] = 0.36$ and
$\Pr[\mathrm{nP}|\mathrm{N}] = 1 - \Pr[\mathrm{P}|\mathrm{N}] = 0.83$.
Also $\Pr[\mathrm{Y}] = 0.7$, $\Pr[\mathrm{N}] = 0.3$.

Sequential decision making: information processing

Example (cont'd)

$$
\begin{aligned}
\mathbb{E}\left[u\left(\mathrm{C}|\mathrm{Y}\right)\right] &= \Pr\left[\mathrm{P}|\mathrm{Y}\right]\left(100-c\right)+\Pr\left[\mathrm{nP}|\mathrm{Y}\right]\left(52.5-c\right) = 82.9-c \\
\mathbb{E}\left[u\left(\mathrm{S}|\mathrm{Y}\right)\right] &= 55-c, \\
\mathbb{E}\left[u\left(\mathrm{C}|\mathrm{N}\right)\right] &= \Pr\left[\mathrm{P}|\mathrm{N}\right]\left(100-c\right)+\Pr\left[\mathrm{nP}|\mathrm{N}\right]\left(52.5-c\right) = 60.58-c \\
\mathbb{E}\left[u\left(\mathrm{S}|\mathrm{N}\right)\right] &= \Pr\left[\mathrm{P}|\mathrm{N}\right]55+\Pr\left[\mathrm{nP}|\mathrm{N}\right]55 = 55-c
\end{aligned}
$$

It is optimal to play C whatever the report says, so bribing gives

$$U(B) = \mathbb{E}\left[u\left(\mathrm{C}|\mathrm{Y}\right)\right]\Pr[\mathrm{Y}] + \mathbb{E}\left[u\left(\mathrm{C}|\mathrm{N}\right)\right]\Pr[\mathrm{N}] = 76.2-c.$$

but direct investment in the company gives $U\left(C\right) = 76.25$, so John decides to discard the opportunity of buying additional information.